

LeRobot Humanoide Build, Break, Repeat

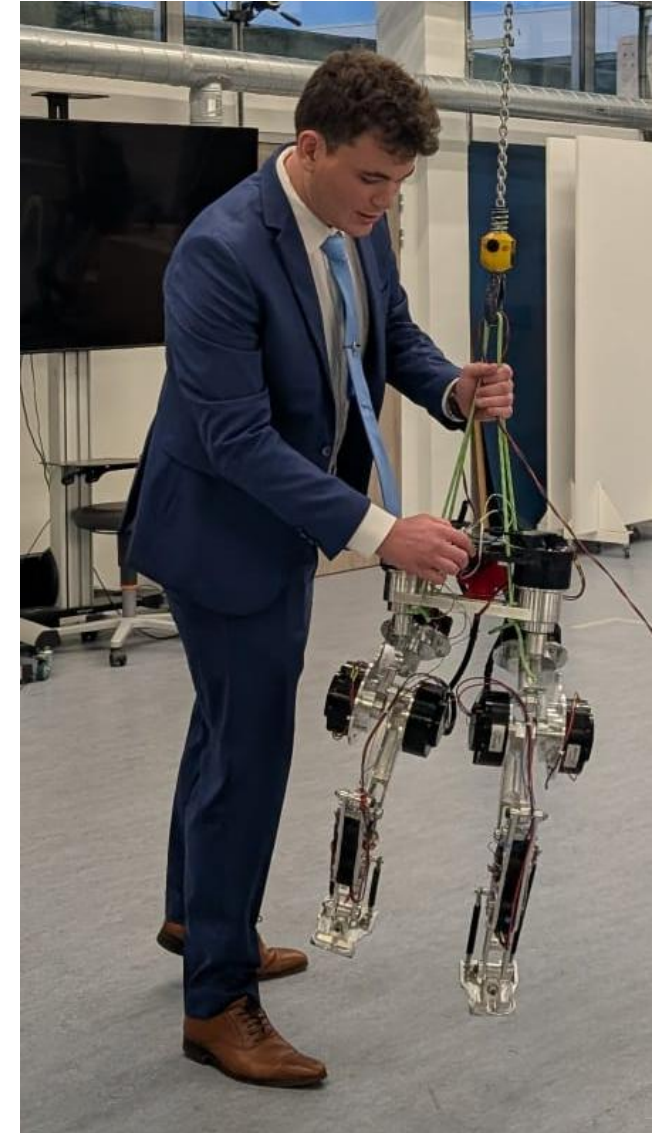
Virgile Batto – 06/05/2026 - Paris



Robotics @ Hugging Face (LeRobot)

- *PhD in legged robot design*
- *hardware → control*

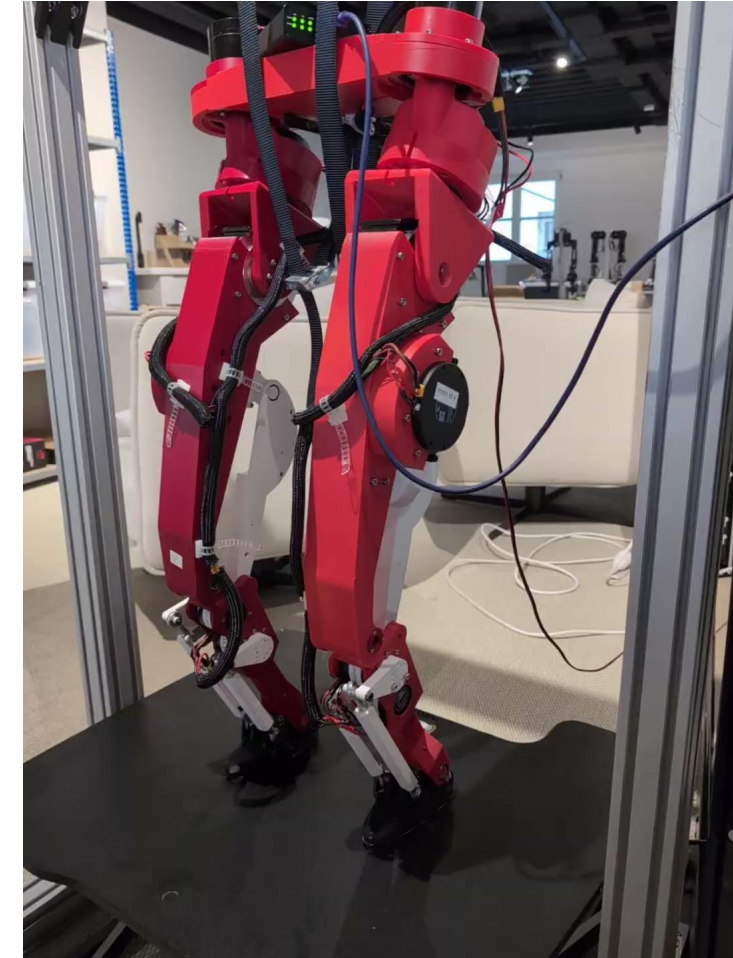
Building robots to experiment in the real world

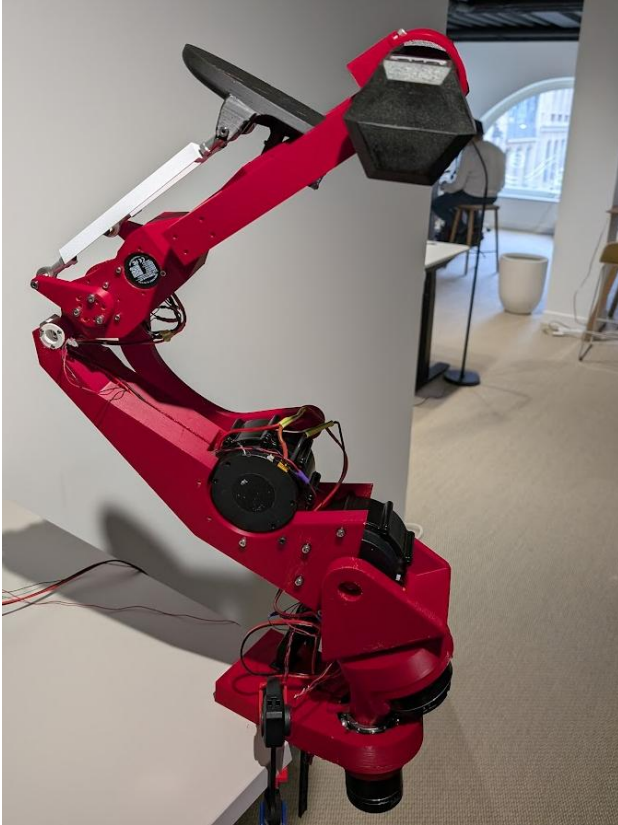


The platform:

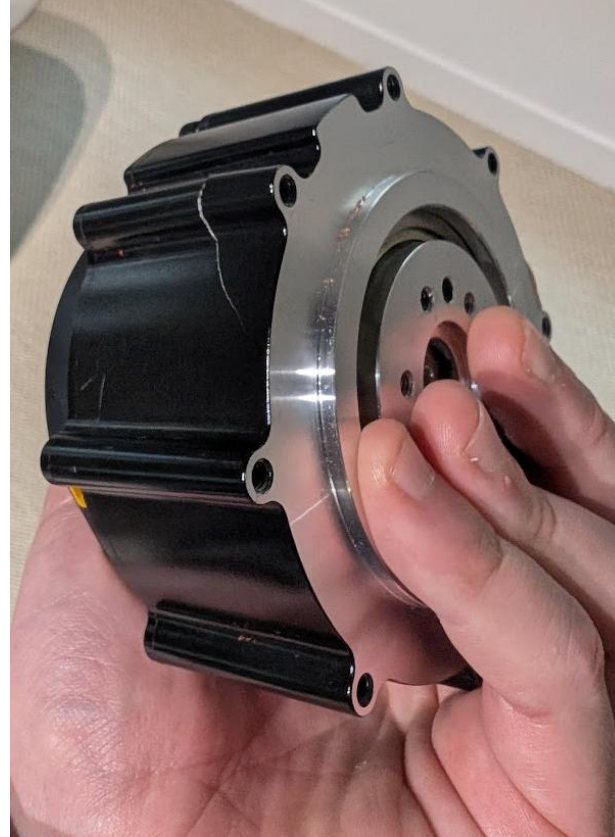
- 70cm tall
- 12 Kg
- ~ 2 500 €
- Built from off-the-shelf parts

Built in weeks, not years

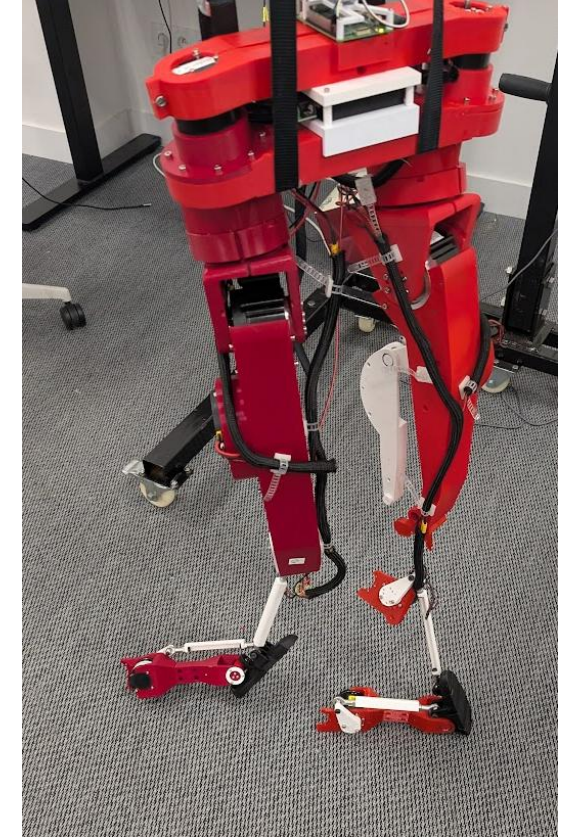




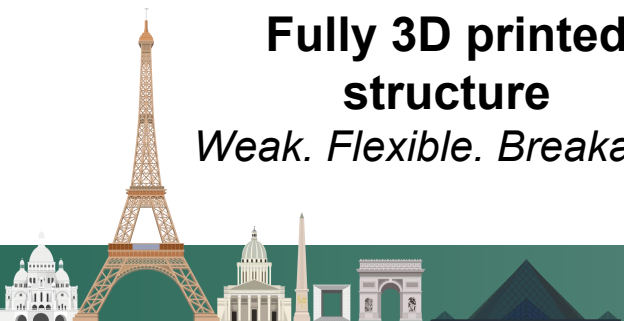
**Fully 3D printed
structure**
Weak. Flexible. Breakable.



Cheap actuators
Low torque. Low precision.

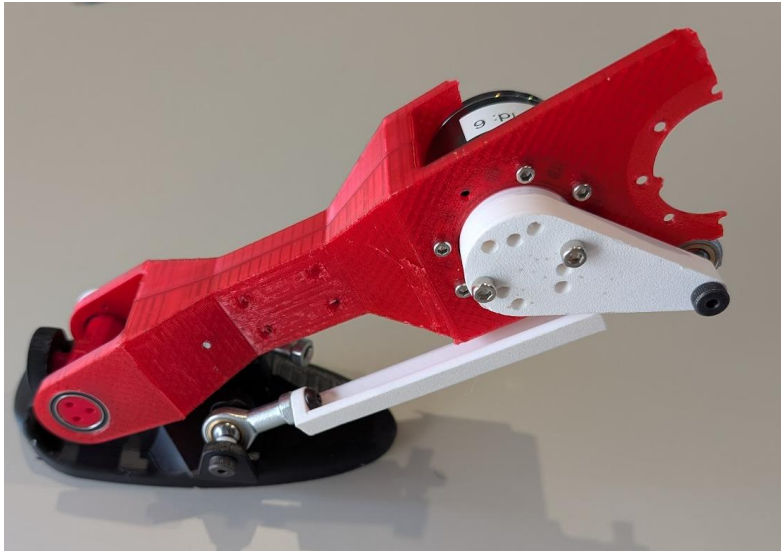


Unstable behavior
Noise. Delay. Chaos.



Fast to iterate ~5 days to reprint everything
Cheap <2500\$ total cost

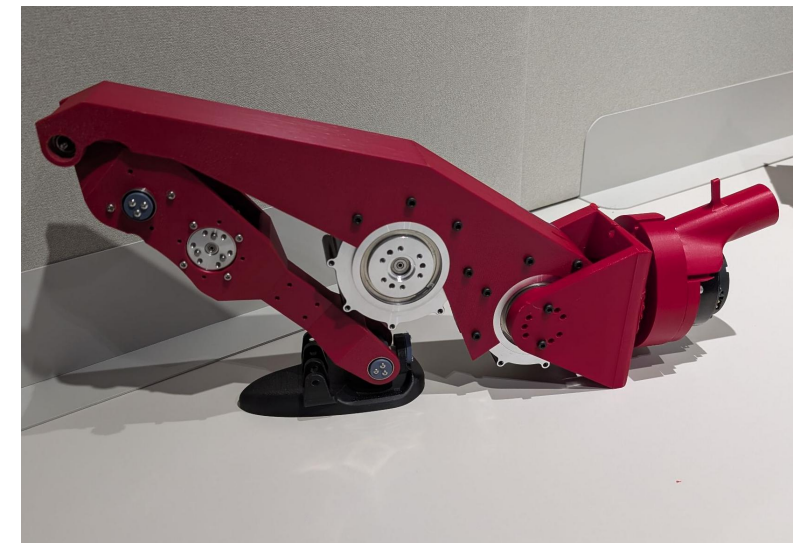
No fear of breaking



Break



Print



Repeat



Open Design

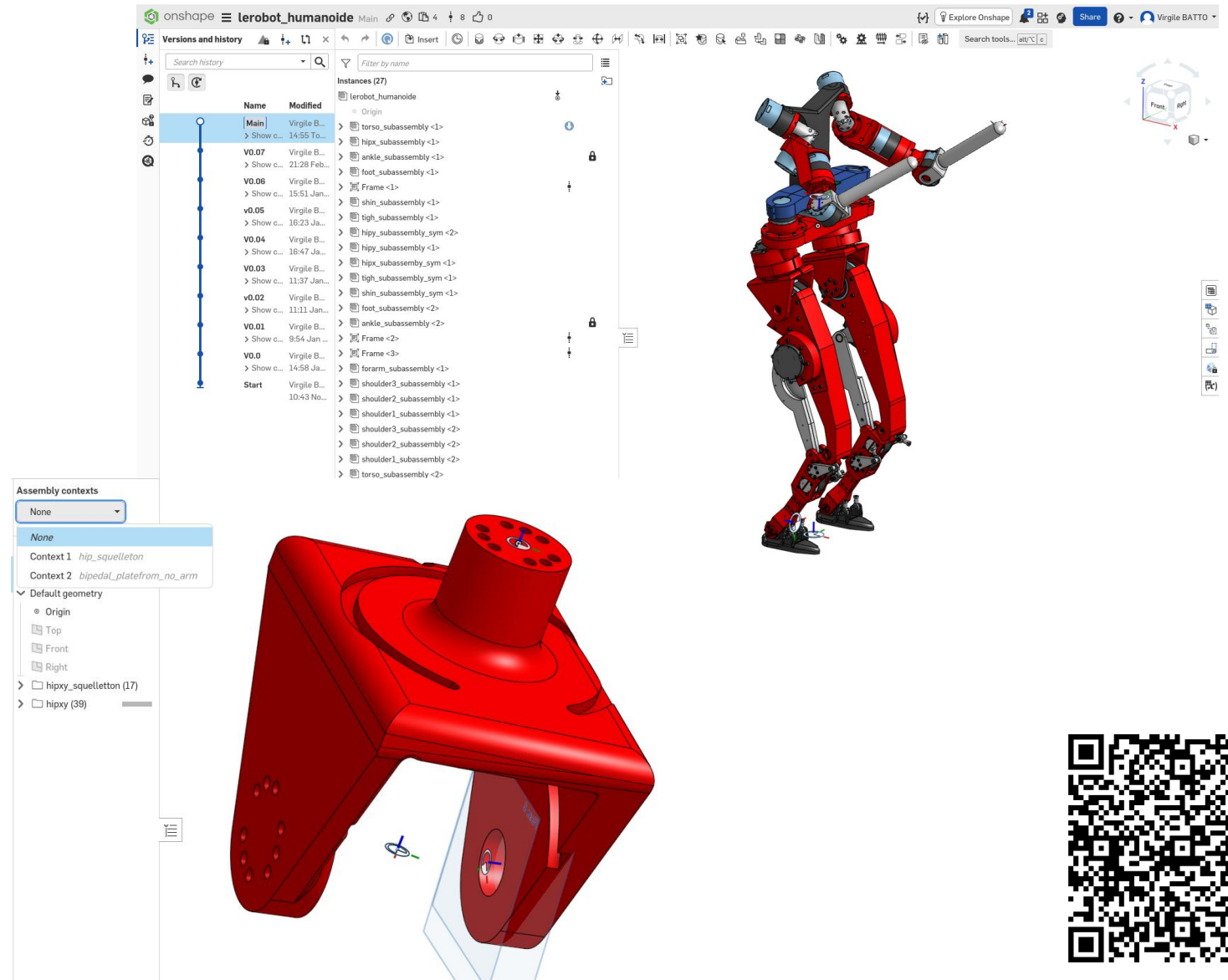
*Fully accessible
fully modifiable*

Fast to modify

Minimal operations

Parametric design (in progress)

Evolving with the robot



Co-design approach

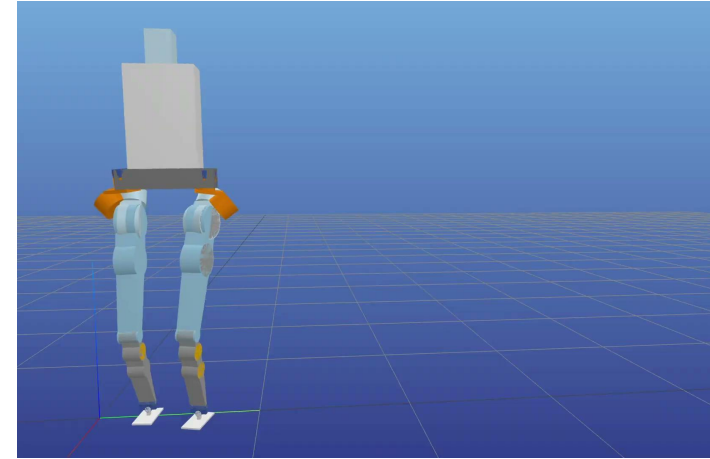
Control informs mechanical design

Geometry matters

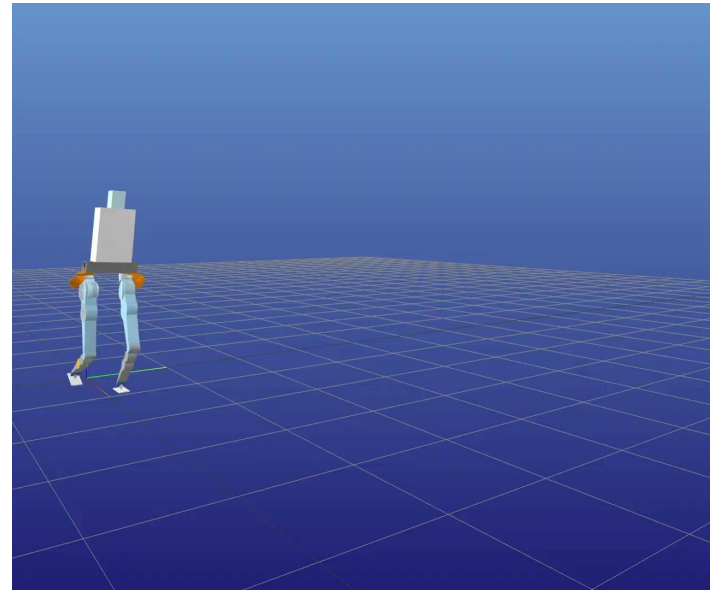
Joint placement impacts torque

Fast iteration loop

Test → simulate → update

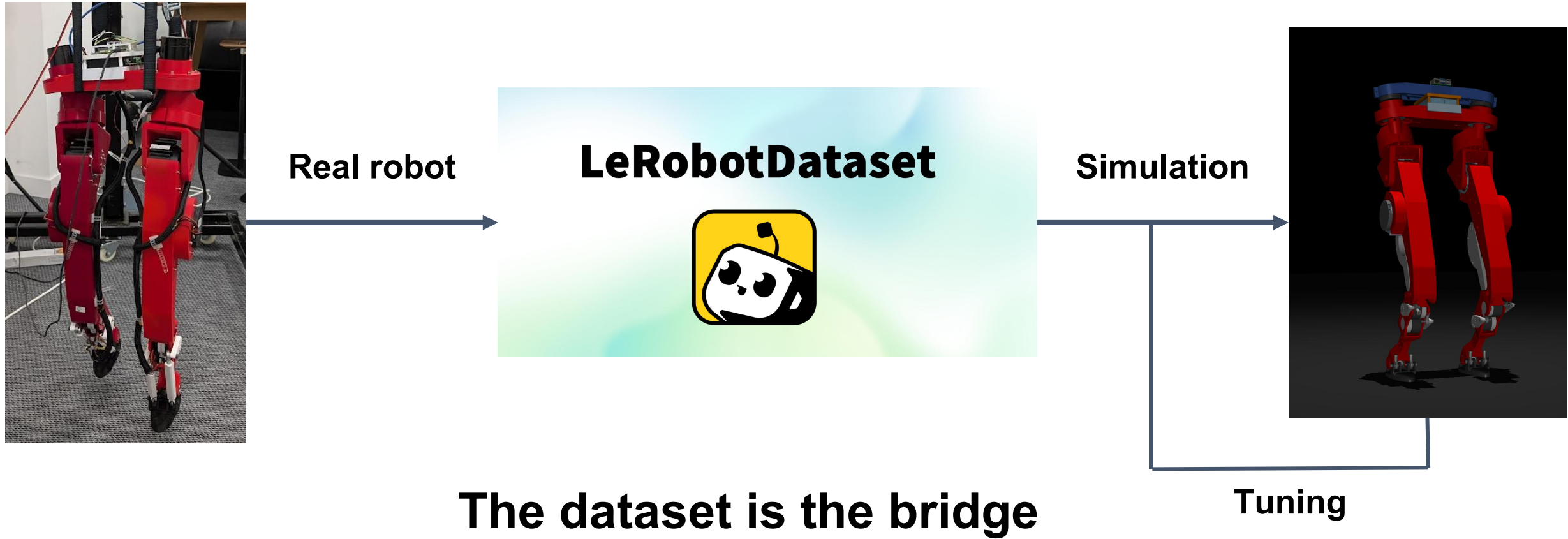


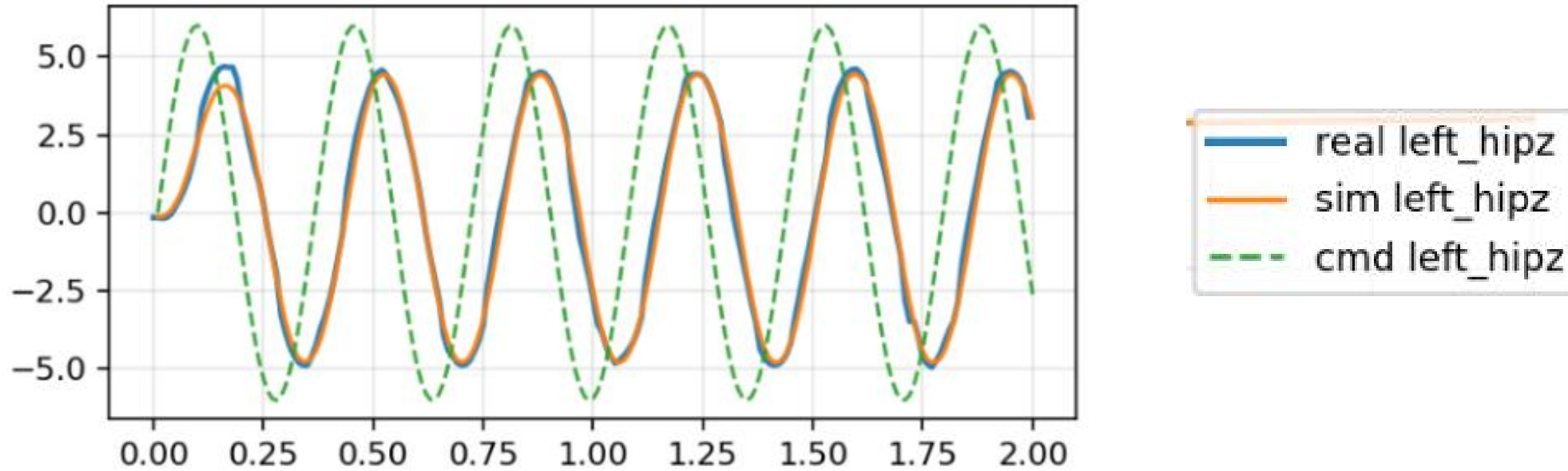
Side walking (OCP)



Forward walking (OCP)



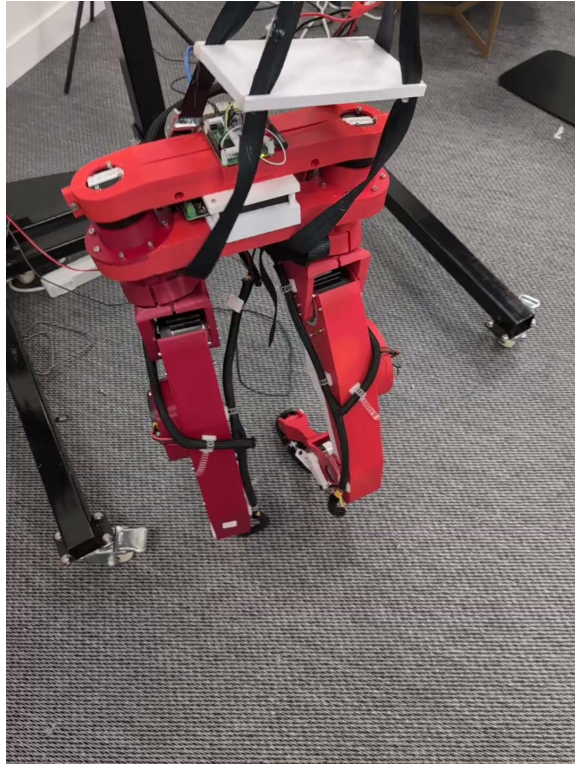




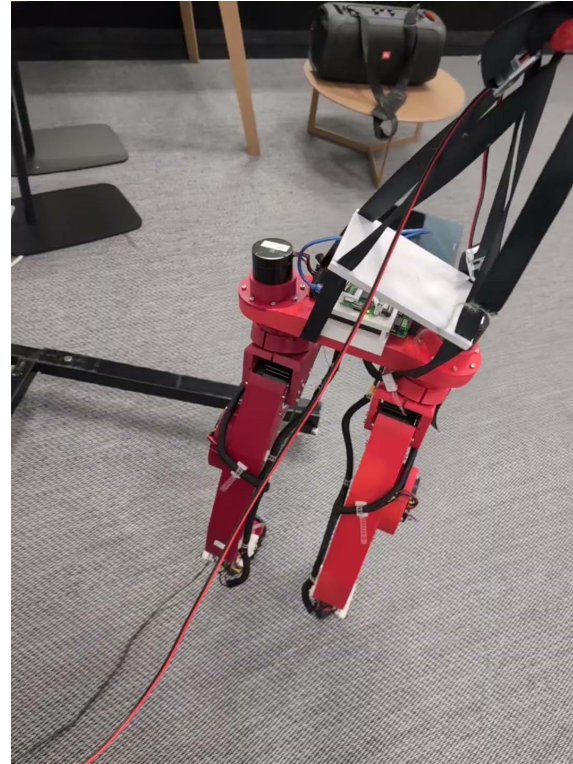
Now simulation behaves like the real robot



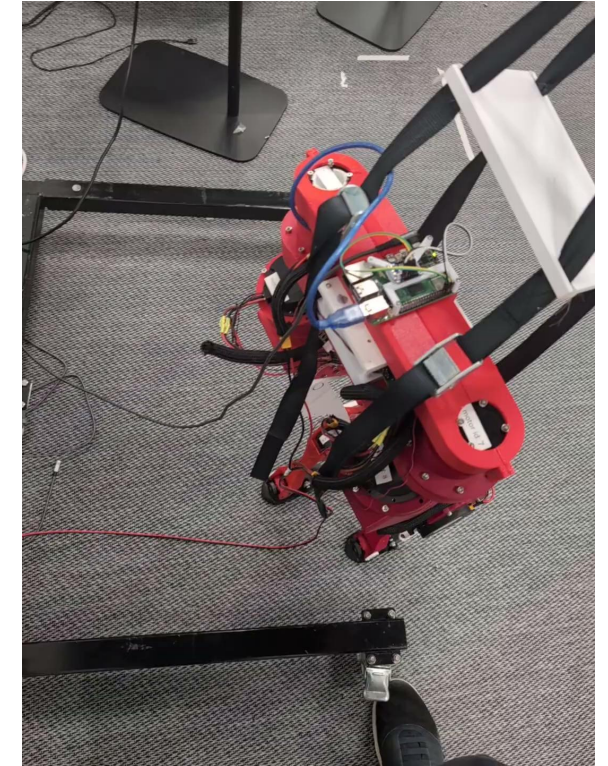
Run policies safely on real hardware



policies A



policies B



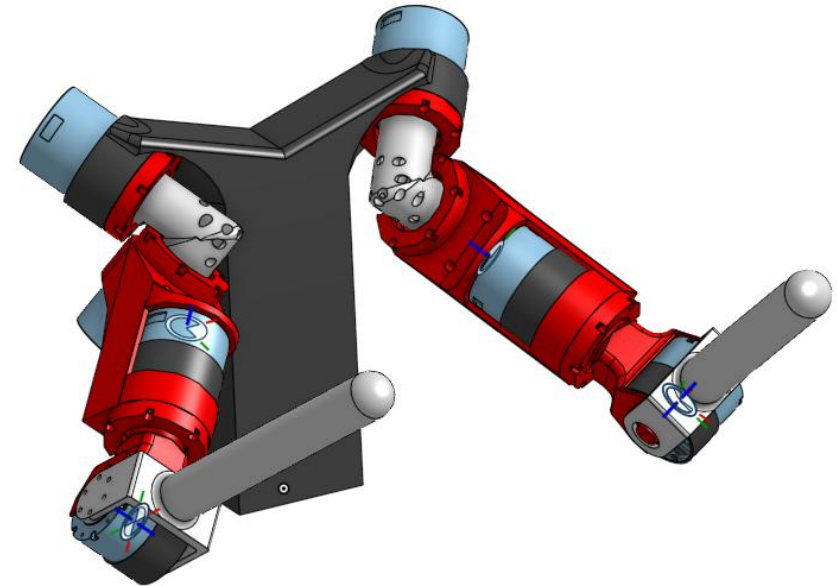
policies C

A platform for real-world experimentation

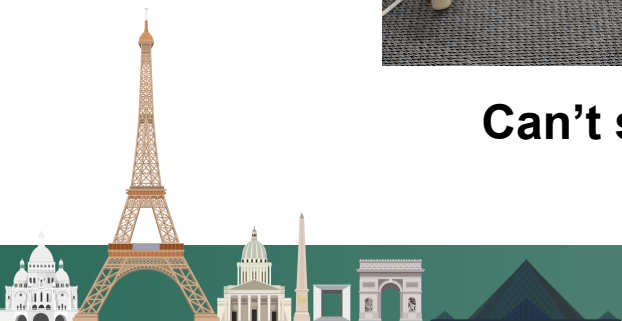




Can't stand alone yet

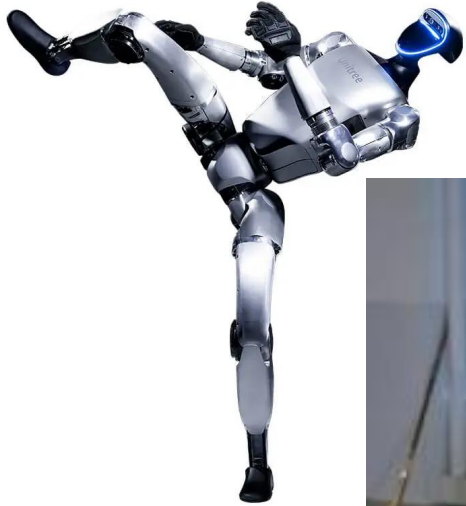


Upper body still missing



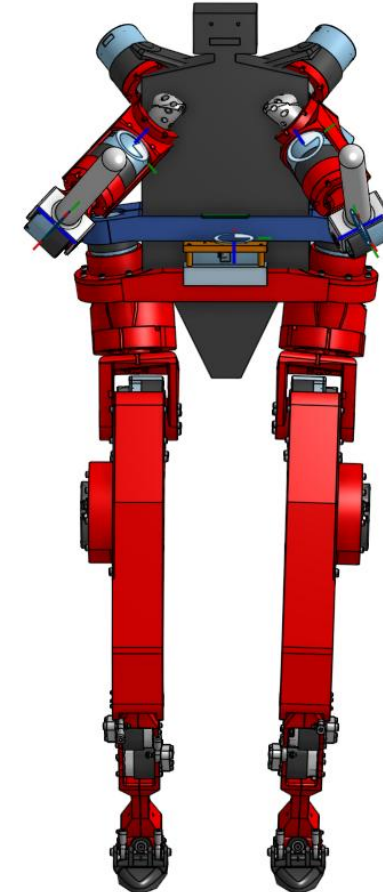
Expensive robots

- *hard to repair*
- *few experiments*



Homemade Robots

- *Easy to repair*
- *Faster learning*



What do we need now ?

GOSIM Paris 2026



